MASOUD ABDOLLAHI, Ph.D.

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PROFESSIONAL SUMMARY

ML Research Scientist with a Ph.D. and 4+ years developing and commercializing production ML systems on wearable and clinical time-series data. Deep expertise in deep learning, time-series analysis, signal processing, and multimodal fusion across IMU, physiological, and sensor streams, the core data modalities behind modern wearable health products. Track record spanning the full model lifecycle: exploratory research, pipeline design, rigorous statistical validation, and production deployment in high-stakes healthcare and real-world environments. **28+ peer-reviewed publications, 700+ citations, and 6 patent filings (1 published).** Currently building clinical AI models for neurological impairment detection on wearable-derived time-series data at Oregon Health & Science University.

CORE TECHNICAL EXPERTISE

- **Wearable & Time-Series ML:** IMU / accelerometer / gyroscope pipelines, physiological waveform analysis, signal processing (filtering, denoising, feature engineering), temporal deep learning (Transformers, RNN-LSTM, 1D-CNN), windowing & segmentation, robust validation on heterogeneous subject populations.
- **Deep Learning Architectures:** Transformers, CNN, RNN-LSTM, self-supervised & semi-supervised learning, transfer learning, multimodal fusion, foundation models, fine-tuning, distillation.
- **Healthcare & Clinical Data:** Wearable sensor time-series, clinical assessments, EMR / clinical report processing, fall-risk and neurological impairment modeling, multi-subject longitudinal studies.
- **Statistical Modeling & Validation:** Hypothesis testing, error estimation, performance monitoring, cross-subject generalization, A/B-style model comparison, calibration, drift detection.
- **Programming & Frameworks:** Python (5+ years), SQL (5+ years), PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, OpenCV, MATLAB, C++.
- **MLOps & Deployment:** End-to-end ML pipelines, large-scale dataset curation, containerized deployment, model monitoring, Azure ML / Azure OpenAI Foundry, reproducible experimentation.

PROFESSIONAL EXPERIENCE

Senior R&D Scientist · Hitachi America, Ltd., Novi, MI *Feb 2023 – Present*

- **Technical leadership: Promoted to Senior R&D Scientist (Jun 2026); lead multiple R&D projects and own a research roadmap.**
- **Full-lifecycle ML commercialization:** Led end-to-end research, training, rigorous validation, and production deployment of a multimodal deep learning system that automated a complex bottleneck process, reducing cycle time by **25%** in a live operational environment; prototype to full rollout with cross-functional scientists, software engineers, product managers, and test engineers.
- **Time-series modeling on wearable sensor data:** Developed multimodal perception systems fusing IMU wearable streams (accelerometer, gyroscope) with computer vision, applying signal processing, feature engineering, and deep temporal models to achieve **>90% classification accuracy** across diverse real-world users.
- **Exploratory research → scalable product features:** Translated proof-of-concept sensing ideas into deployed, commercialized features; characterized novel signal-derived features through data exploration, ablation studies, and statistical validation.
- **Transformer models for structured health-style data:** Designed and built Transformer-based models on structured operational and report data for classification, anomaly detection, and quality assessment; pipelines directly analogous to clinical NLP over EMRs and structured health records.
- **Rigorous statistical analysis & monitoring:** Built evaluation frameworks with error estimation, subgroup performance analysis, and production monitoring to ensure robustness across heterogeneous populations and meet strict accuracy requirements for deployment.
- **Large-scale real-world data pipelines:** Collected, curated, and managed 500+ hours/year of deployment data; built reproducible training and evaluation pipelines from heterogeneous sensor and vision streams.
- **Cross-functional communication & IP:** Presented technical findings to leadership, external partners, and non-technical stakeholders; filed 6 patents in AI/ML, generative AI, and sensing (inventor on all; 1 published, 2 under USPTO review, 3 approved for filing).

- **AI Data Challenge winner:** Won the 2026 Hitachi America R&D AI Data Challenge with a physics-informed, data-efficient model that forecasts day-ahead hourly electricity prices at all 243 buses of a power-grid benchmark from only 30 labeled days.

Research Engineer III (Part-time) · Oregon Health & Science University, Remote *Feb 2024 – Present*

- **Wearable-sensor AI for Parkinson's disease:** Developing sensor-based deep learning models to analyze freezing of gait and fall risk in Parkinson's disease, applying time-series ML and signal processing to wearable-derived data in a research-to-deployment pipeline alongside clinicians and medical domain experts.
- **Collaboration with clinical domain experts:** Bridging quantitative ML modeling with clinical knowledge to inform patient-centered model design, validation strategy, and interpretation of health-relevant signals.

R&D Scientist, Internship · Aster Pet Sciences *Jun 2021 – Aug 2021*

- **End-to-end ownership of an AI project:** Led data collection, feature engineering, model design, and shipment of a real-time IMU-based activity recognition system (SVM, >70% accuracy), independently developed and deployed.

EDUCATION

Ph.D., Mechanical and Industrial Engineering · Rochester Institute of Technology 2017 – 2023

Dissertation: Machine Learning for Fall Risk Assessment: integrating clinical tests, wearable motion sensors, and multimodal time-series data; a direct precursor to building ML systems for wearable health.

M.Sc., Mechanical Engineering · Sharif University of Technology 2012 – 2015

SELECTED PUBLICATIONS

28+ peer-reviewed publications · 700+ citations · Full list on Google Scholar

- Abdollahi, M., Arcobelli, V. A., Mancini, M., Horak, F. B., et al. (2026). *360° turning in individuals with mild to moderate Parkinson's disease: a clustering analysis of segmental coordination strategies.* Journal of NeuroEngineering and Rehabilitation.
- Abdollahi, M., Zhou, Q., & Yuan, W. (2024). *Smart wearable insoles in industrial environments: A systematic review.* Applied Ergonomics 118, 104250.
- Abdollahi, M., Rashedi, E., et al. (2024). *Fall risk assessment in stroke survivors: a machine learning model using detailed motion data.* Sensors 24(3), 812.
- Jahangiri, S., Abdollahi, M., et al. (2024). *An inpatient fall risk assessment tool: Application of ML models on intrinsic and extrinsic risk factors.* Machine Learning with Applications 15, 100519.
- Jahangiri, S., Abdollahi, M., et al. (2024). *A machine learning model to predict heart failure readmission: toward optimal feature set.* Frontiers in Artificial Intelligence 7, 1363226.
- Karvekar, S., Abdollahi, M., & Rashedi, E. (2021). *Smartphone-based human fatigue level detection using machine learning.* Ergonomics, 1–28.

PATENTS

- Zhou, Q., & Abdollahi, M. Smart Sole Architecture for Measuring Ground Reaction Force and Method for Unifying Coordinates for Feet Positions and Gestures. U.S. Patent Application Publication US 2026/0198641 A1. Published Jul. 16, 2026.
- Abdollahi, M., et al. Agentic AI & generative AI system for industrial operations. U.S. Patent Application. Under USPTO Review. (2025).
- Abdollahi, M., et al. Sensor-based human performance assessment system. U.S. Patent Application. Under USPTO Review. (2026).
- Abdollahi, M., et al. Agentic AI, generative AI, sensor-based human performance assessment & training (3 applications). U.S. Patent Applications. Approved for filing, in preparation. (2026).